

Getting Started with PicoClaw on Raspberry Pi

A Beginner's Guide for Students

APWE Singapore

Why Use a Raspberry Pi for PicoClaw?

You might be wondering — *"Why not just run PicoClaw on my laptop or school computer?"* Great question! Here are some reasons we're using a **Raspberry Pi** instead:

Safety and Privacy

When PicoClaw runs on a Raspberry Pi, all your conversations with the AI stay **on the device** — they don't travel across the school network or get stored on a shared laptop. This makes it a **safer, more private** environment for young learners to experiment with AI.

A Dedicated Learning Sandbox

The Raspberry Pi is a **self-contained mini-computer** just for PicoClaw. You can explore, make mistakes, and even break things — without affecting their personal laptops, school-issued devices, or important schoolwork. If something goes wrong, we simply reflash the SD card and start fresh.

What You'll Need

Hardware Checklist:

- Raspberry Pi
- MicroSD card (32GB+)
- Power adapter, mouse, keyboard, monitor with HDMI cable
- Internet connection (Wi-Fi or Ethernet)

Accounts You'll Need:

- A Google account (for Gemini API key)
- A Telegram account (for the bot)

Setting Up Your Raspberry Pi

Flash the SD Card

1. Download **Raspberry Pi Imager** from the [official website](#).
2. Insert your microSD card into your computer.
3. Open Imager → Choose **Raspberry Pi OS (64-bit)** → Choose your SD card → Click **Write**.



First Boot

1. Insert SD card into Raspberry Pi.
2. Connect monitor, keyboard, mouse, and power.
3. Follow on-screen setup wizard (language, Wi-Fi, password).

Connect to Wi-Fi

- Click the Wi-Fi icon (top-right) → choose network → enter password.

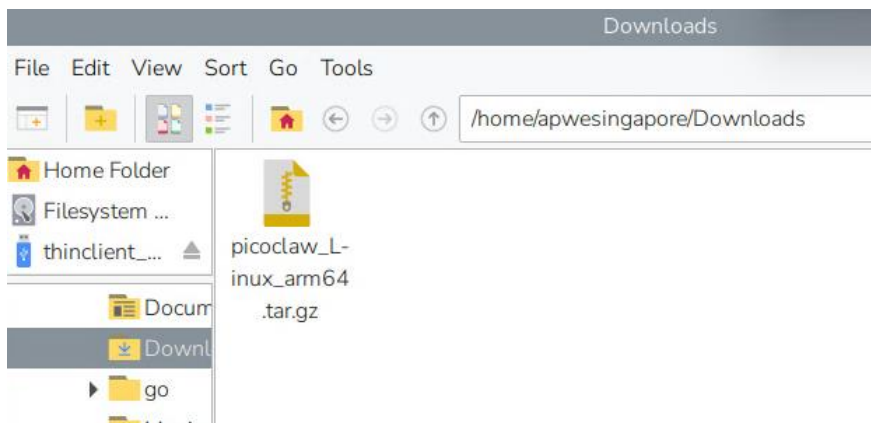


Installing PicoClaw

Download PicoClaw

1. Open the Chromium browser from the taskbar on your Raspberry Pi.
2. In the address bar, type: <https://picoclaw.io/download/> and press Enter.
3. On the download page, find the section labeled Linux ARM 64.
4. Click the Linux ARM 64 download button.

5. When prompted, click Save (or Keep) to download the file.
6. The file (for example, picoclaw-linux-arm64.tar.gz) will be saved to your Downloads folder.



💡 Why ARM 64? Raspberry Pi 4 and 5 use a 64-bit ARM processor, so this is the correct version for your device.

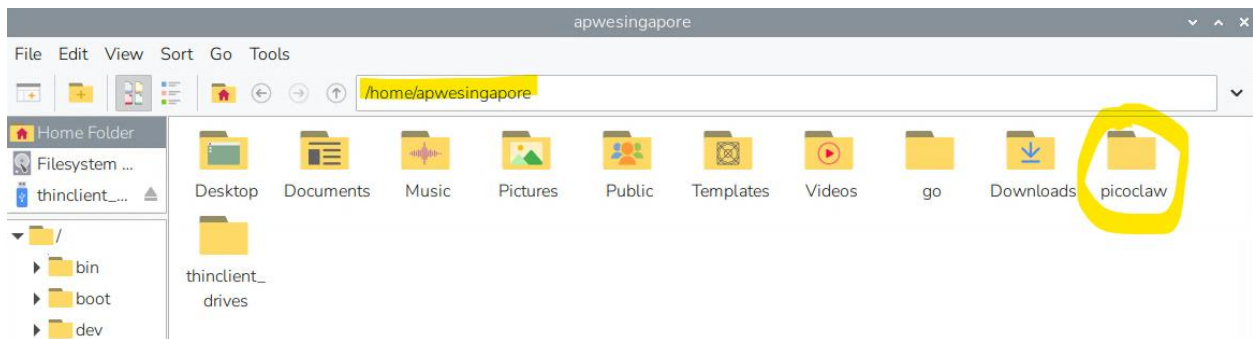
Create a PicoClaw Folder and Unpack the File

To keep your files organised, we'll create a new folder called `picoclaw` inside your Home folder. Then we'll move the downloaded file there and unpack it.

Part A — Create the PicoClaw Folder

1. Open File Manager from the taskbar (folder icon).
2. In the left sidebar, click Home (the house icon) to go to `/home/apwesingapore`.

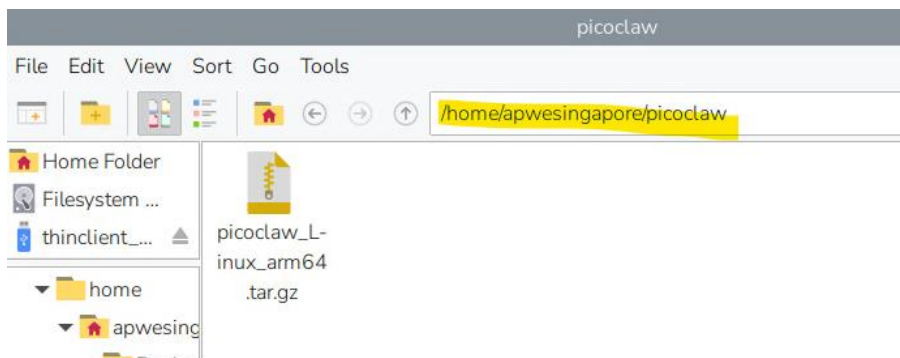
3. Right-click on any empty space inside the Home folder.
4. From the menu, choose Create New → Folder (or New Folder).
5. A small dialog will appear asking for a name. Type picoclaw (all lowercase).
6. Click OK (or press Enter).
7. A new folder named picoclaw will appear inside your Home folder.



💡 Tip — Check the path: At the top of File Manager, you should see the address bar showing /home/apwesingapore. If you don't see the address bar, press Ctrl + L to reveal it.

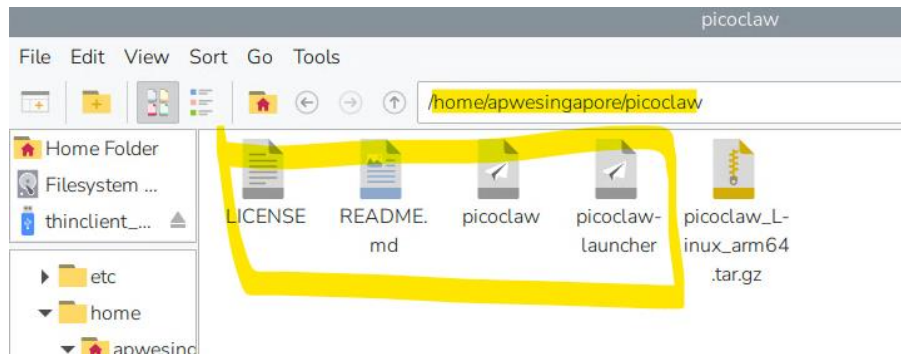
Move the Downloaded File Into the PicoClaw Folder

1. In the left sidebar of File Manager, click Downloads.
2. Find the file picoclaw-linux-arm64.tar.gz (or similar name).
3. Right-click the file → choose Cut (or press Ctrl + X).
4. In the left sidebar, click Home to return to /home/apwesingapore.
5. Double-click the picoclaw folder to open it.
6. Right-click on any empty space inside the folder → choose Paste (or press Ctrl + V).
7. The .tar.gz file should now appear inside /home/apwesingapore/picoclaw.



Unpack the File

1. Make sure you are inside the picoclaw folder (/home/apwesingapore/picoclaw).
2. Right-click the file picoclaw-linux-arm64.tar.gz.
3. From the menu, select Extract Here.
4. Wait a few seconds while the file unpacks.

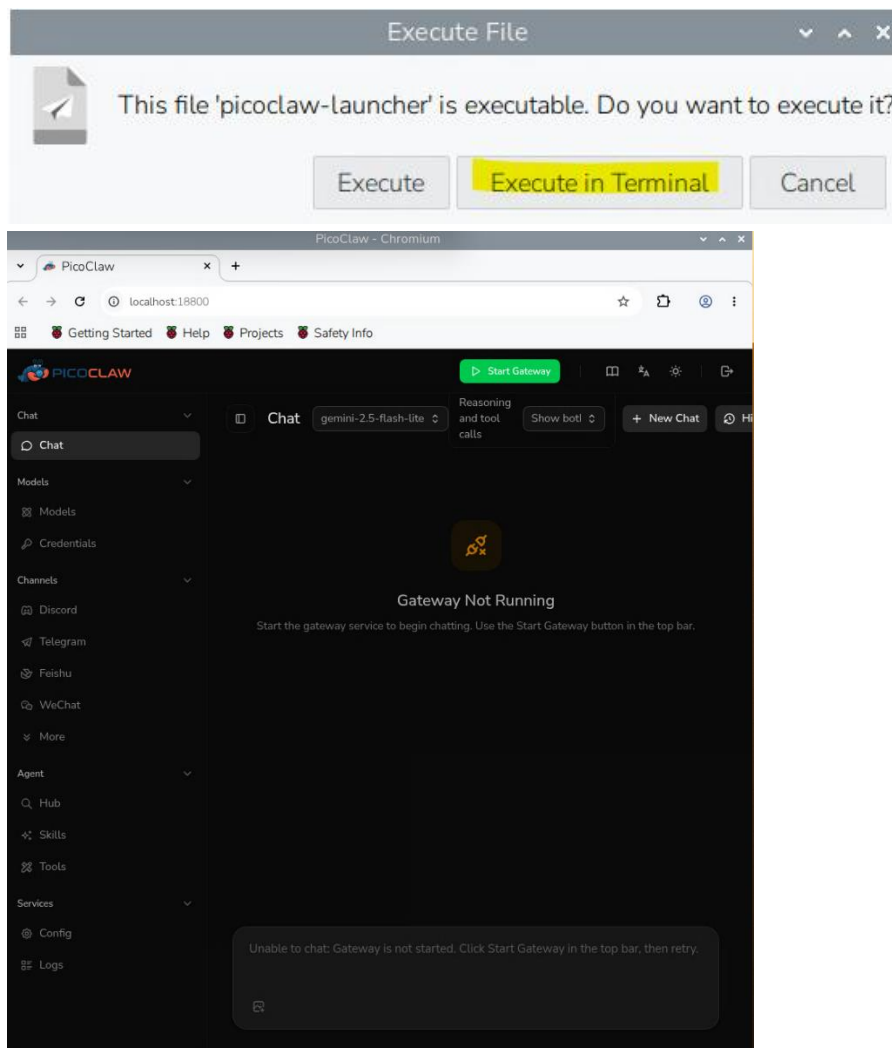


Launch PicoClaw

The PicoClaw folder includes a ready-made launcher file. Just double-click it, and the PicoClaw dashboard will open automatically in your web browser.

Step-by-Step

1. Find the file named picoclaw-launcher.
2. Double-click the picoclaw-launcher file.
3. A dialog will appear asking what to do — choose Execute in Terminal.
4. A black terminal window will open and show some start-up messages. Leave this window open — it's running PicoClaw in the background.
5. After a few seconds, Chromium will open automatically and display the PicoClaw dashboard.



✅ Checkpoint: If the dashboard loads in Chromium, you're ready to go! Continue to next section to connect PicoClaw to an LLM.

⚠️ Important — Keep the Terminal Window Open

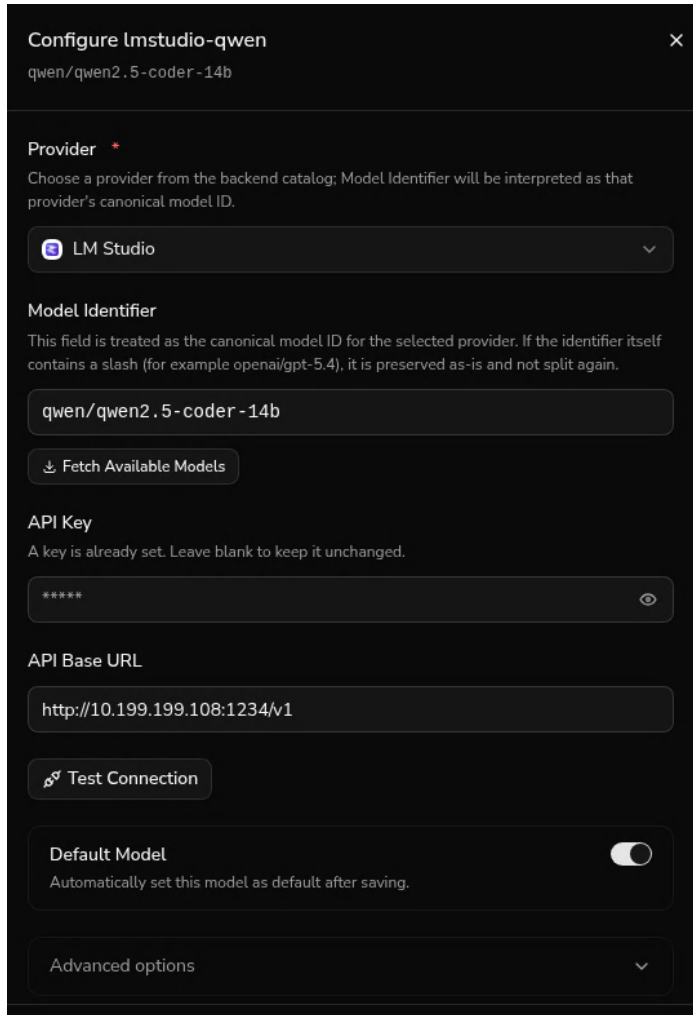
The black terminal window is what keeps PicoClaw running. If you close it, the dashboard in Chromium will stop working.

- To use PicoClaw: Leave the terminal window minimised in the background.
- To stop PicoClaw: Close the terminal window (or press Ctrl + C inside it).

Connecting PicoClaw to an LLM

Option A — Qwen/Llama (Hosted by Us)

1. In PicoClaw: go to **Models** → **Add Model** and fill in as follows.



Configure lmstudio-qwen ×

qwen/qwen2.5-coder-14b

Provider *

Choose a provider from the backend catalog; Model Identifier will be interpreted as that provider's canonical model ID.

LM Studio

Model Identifier

This field is treated as the canonical model ID for the selected provider. If the identifier itself contains a slash (for example openai/gpt-5.4), it is preserved as-is and not split again.

qwen/qwen2.5-coder-14b

↓ Fetch Available Models

API Key

A key is already set. Leave blank to keep it unchanged.

API Base URL

http://10.199.199.108:1234/v1

🔧 Test Connection

Default Model ☒

Automatically set this model as default after saving.

Advanced options

✅ **Use this if:** You want privacy.

Option B — Google Gemini (Cloud)

2. Open browser → visit [Google AI Studio](#).
3. Sign in with Google → click **Create API Key** → copy the key.
4. In PicoClaw: go to **Models** → **Add Model** and fill in as follows.

Configure gemini-2.5-flash-lite

gemini-2.5-flash-lite

Provider

Choose a provider from the backend catalog; Model Identifier will be interpreted as that provider's canonical model ID.

OpenAI

Model Identifier

This field is treated as the canonical model ID for the selected provider. If the identifier itself contains a slash (for example openai/gpt-5.4), it is preserved as-is and not split again.

gemini-2.5-flash-lite

gpt-5.4

gpt-5.4-mini

gpt-5.5

Fetch Available Models

API Key

A key is already set. Leave blank to keep it unchanged.

AI****W8

API Base URL

https://generativelanguage.googleapis.com/v1beta/openai

Test Connection

Default Model

Automatically set this model as default after saving.

Advanced options

Cancel

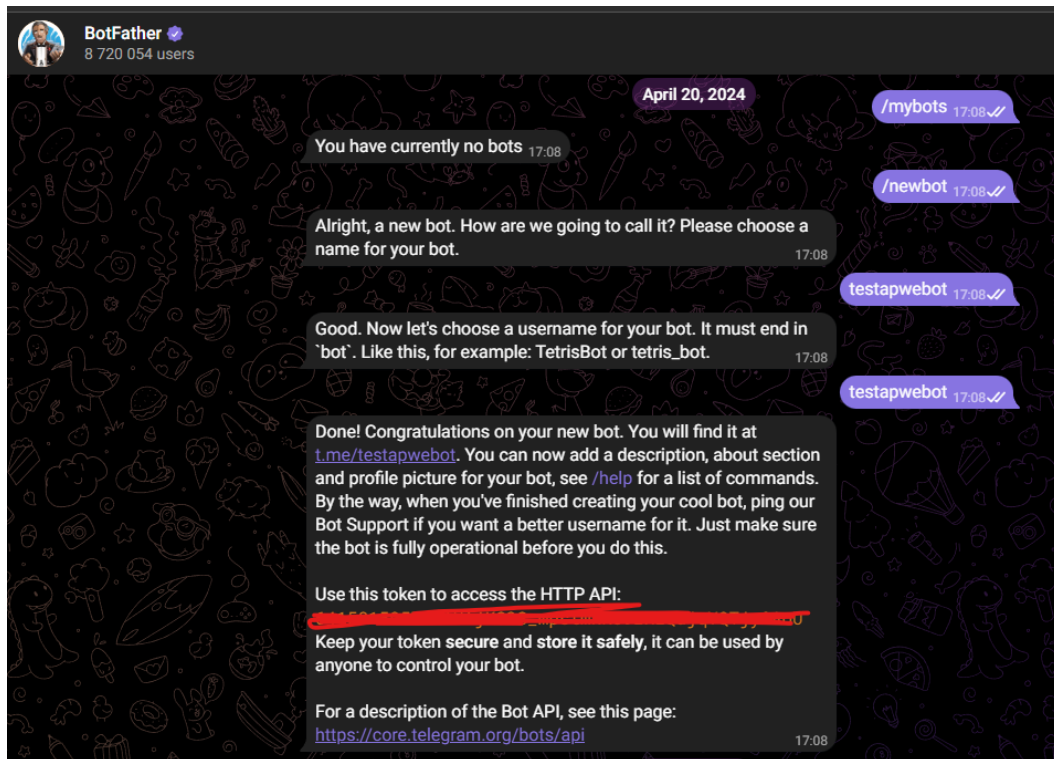
Save

✔ **Use this if:** You want faster, more powerful responses.

Connecting to a Telegram Bot

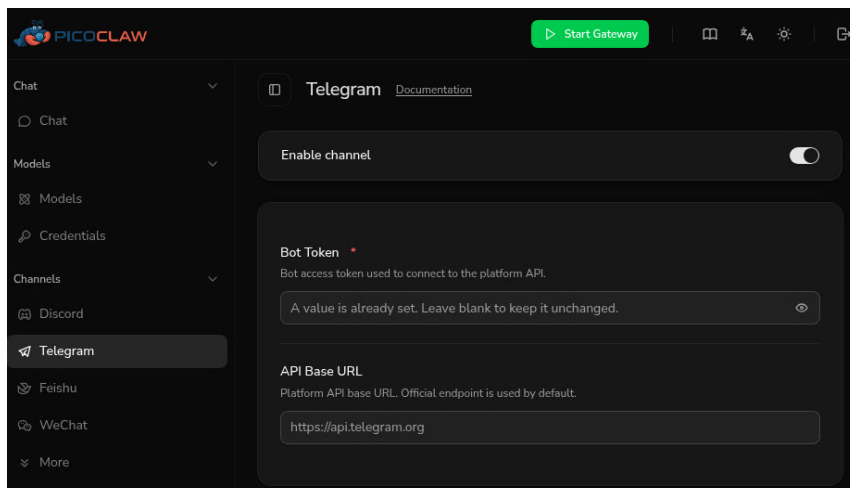
Create a Bot in Telegram

1. Open Telegram on your phone → search for **@BotFather**.
2. Type `/newbot` → follow prompts (give your bot a name + username).
3. Copy the **API token** BotFather gives you.



Link Bot to PicoClaw

1. In PicoClaw: go to **Telegram**.
2. Click Enable channel
3. Paste the **API token** → click **Connect**.
4. Click Start Gateway
5. Start the bot with **/start**, and then after that you can chat with your AI agent through Telegram bot.



Test the Bot

- On your phone, send a message to your bot. You should get an AI-powered response!

